UNIVERSAL OPENNESS OF A REPRESENTABLE MORPHISM OF ALGEBRAIC SPACES TESTED BY BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof shares its mechanism with the closed and submersive criteria; the author writes it out separately and it is counted once here.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Description of the points of a fibre product of algebraic spaces. Properties of Spaces, Section spaces-properties-section-points, used at proof lines 936-947. Gives the square of topological spaces, its open surjective horizontal arrows and the surjective comparison map.
- Universal openness of a representable morphism in the functorial sense. Section section-representable, used at proof lines 922-926 and 933-935. Fixes condition (1) and gives openness of V x_Y X -> V.
- Openness detected after an open surjective comparison. used at proof lines 944-950. Transfers openness from the left to the right vertical arrow.

Source reference. Original section 03HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4559–4559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 03BT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4360–4360. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a representable morphism of algebraic spaces over S. The following are equivalent*

(1) *f is universally open (in the sense of Section 03HA), and*
(2) *for every morphism of algebraic spaces $Z \to Y$ the morphism of topological spaces $|Z \times_Y X| \to |Z|$ is open.*

Proof. Assume (1), and let $Z \to Y$ be as in (2). Choose a scheme V and a surjective étale morphism $V \to Y$. By assumption the morphism of schemes $V \times_Y X \to V$

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is universally open. By Properties of Spaces, Section 03BT in the commutative diagram

$$\begin{array}{ccc} |V \times_Y X| & \longrightarrow & |Z \times_Y X| \\ \downarrow & & \downarrow \\ |V| & \longrightarrow & |Z| \end{array}$$

the horizontal arrows are open and surjective, and moreover

$$|V \times_Y X| \longrightarrow |V| \times_{|Z|} |Z \times_Y X|$$

is surjective. Hence as the left vertical arrow is open it follows that the right vertical arrow is open. This proves (2). The implication (2) $\Rightarrow$ (1) is immediate from the definitions. $\square$